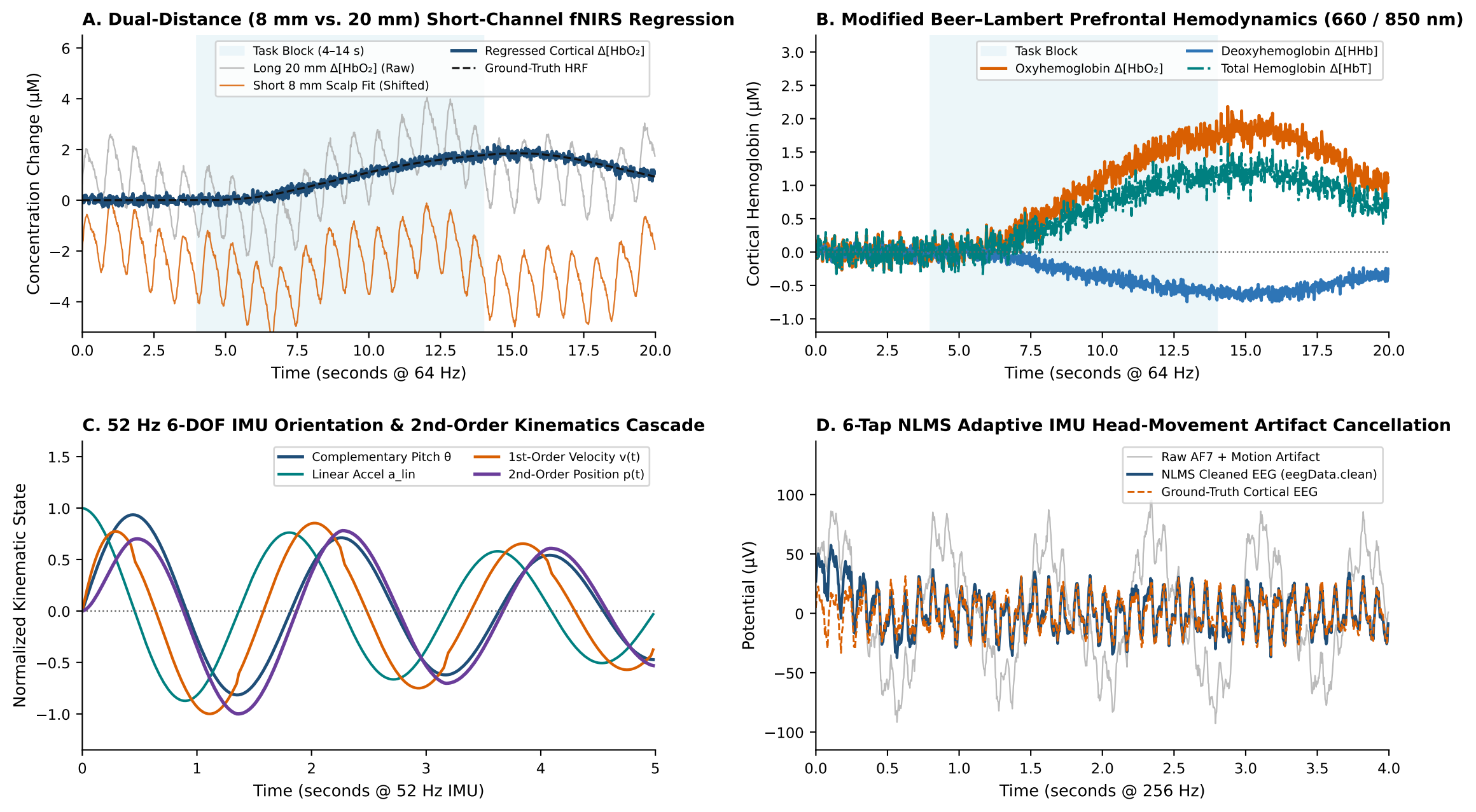

A. Dual-Distance (8 mm vs. 20 mm) Short-Channel fNIRS Regression
Task Block (4–14 s)
Long 20 mm Δ[HbO₂] (Raw)
Short 8 mm Scalp Fit (Shifted)
Regressed Cortical Δ[HbO₂]
Ground-Truth HRF
Concentration Change (μM)
Time (seconds @ 64 Hz)
B. Modified Beer–Lambert Prefrontal Hemodynamics (660 / 850 nm)
Task Block
Oxyhemoglobin Δ[HbO₂]
Deoxyhemoglobin Δ[HHb]
Total Hemoglobin Δ[HbT]
Cortical Hemoglobin (μM)
Time (seconds @ 64 Hz)
C. 52 Hz 6-DOF IMU Orientation & 2nd-Order Kinematics Cascade
Complementary Pitch θ
Linear Accel a_lin
1st-Order Velocity v(t)
2nd-Order Position p(t)
Normalized Kinematic State
Time (seconds @ 52 Hz IMU)
D. 6-Tap NLMS Adaptive IMU Head-Movement Artifact Cancellation
Raw AF7 + Motion Artifact
NLMS Cleaned EEG (eegData.clean)
Ground-Truth Cortical EEG
Potential (μV)
Time (seconds @ 256 Hz)